

MEDISL

Smart Hospital Queue & Appointment Management System

Project Proposal

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1. Executive Summary

The healthcare sector in Sri Lanka faces significant challenges related to overcrowding, long waiting times, inefficient patient flow management, and limited access to healthcare information. Many government hospitals still rely on manual queue management systems, causing inconvenience for patients and administrative difficulties for healthcare staff.

MEDISL (Medical Smart Lanka) is a multilingual Smart Hospital Queue and Appointment Management System designed to modernize hospital operations through digital transformation. The system enables patients to schedule appointments online, monitor queue status in real time, receive notifications, and access healthcare services more efficiently. The platform supports Sinhala, Tamil, and English languages to ensure accessibility for all citizens.

The proposed solution aims to improve patient satisfaction, reduce waiting times, enhance healthcare service delivery, and optimize hospital resource utilization.

2. Introduction

Healthcare institutions play a vital role in society; however, many hospitals in Sri Lanka continue to struggle with outdated appointment and queue management processes. Patients often spend several hours waiting to meet doctors, while hospital staff face challenges in managing large patient volumes.

Advancements in Information and Communication Technology (ICT) provide opportunities to improve healthcare operations through digital systems. By introducing a smart queue and appointment management platform, hospitals can streamline patient flow, provide real-time information, and improve service quality.

MEDISL seeks to address these issues by offering an integrated digital platform that facilitates appointment booking, queue tracking, multilingual communication, and administrative management.

3. Problem Statement

Most government hospitals in Sri Lanka currently operate using manual registration and queue management methods. These traditional systems create several challenges:

- Long patient waiting times ranging from 3 to 8 hours.
- Lack of visibility regarding queue position and waiting duration.
- Overcrowding in waiting areas.
- Inefficient patient flow management.
- Difficulty managing priority patients such as elderly individuals and pregnant women.
- Language barriers affecting communication.
- Limited appointment scheduling capabilities.

Patients currently have no reliable method to:

- Book appointments remotely.
- Monitor queue status.
- Receive notifications regarding appointment updates.
- Access healthcare information in their preferred language.

As a result, hospitals experience reduced efficiency while patients encounter frustration and inconvenience.

4. Project Aim

To design and develop a multilingual Smart Hospital Queue and Appointment Management System that improves healthcare service delivery by reducing waiting times, enhancing patient experience, and optimizing hospital operations.

5. Project Objectives

Main Objective

To create a digital platform that enables efficient appointment scheduling and real-time queue management for hospitals in Sri Lanka.

Specific Objectives

The system aims to:

- Provide online appointment booking facilities.
- Enable real-time queue tracking.
- Send automated notifications through SMS and mobile applications.
- Support Sinhala, Tamil, and English languages.
- Prioritize elderly, disabled, and pregnant patients.
- Improve hospital workflow management.
- Reduce patient waiting times.
- Enhance communication between patients and healthcare providers.
- Generate reports and analytics for hospital administrators.

6. Scope of the Project

The project covers the following functionalities:

Included Scope

- Patient registration and authentication.

- Doctor schedule management.
- Appointment booking and cancellation.
- Real-time queue monitoring.
- Notification management.
- Multilingual user interface.
- Administrative dashboard.
- Patient prioritization system.
- Hospital performance reporting.

Excluded Scope

- Electronic Medical Records (EMR).
- Online payment processing.
- Pharmacy management.
- Laboratory management systems.
- National healthcare database integration

7. Target Users

The system is designed for the following user groups:

Patients

Patients can:

- Register and create accounts.
- Book appointments.
- Monitor queue positions.
- Receive notifications.
- Select preferred language.

Doctors

Doctors can:

- View appointment schedules.
- Access patient queue information.
- Update consultation status.
- Manage delays and availability.

Hospital Administrators

Administrators can:

- Manage doctor profiles.

- Configure schedules.
- Monitor queue operations.
- Generate reports.
- Manage patient priorities.

8. Proposed Solution

MEDISL will provide a centralized digital platform accessible via web and mobile devices.

The system workflow is as follows:

- Patient registers and logs in.
- Patient selects preferred language.
- Patient books an appointment with a doctor.
- System generates a queue number.
- Queue updates are displayed in real time.
- Notifications are sent before the patient's turn.
- Doctor updates consultation status.
- Administrative staff-monitor and manage operations.

9. Functional Requirements

Patient Module

- User registration.
- Secure login and authentication.
- Appointment booking.
- Appointment cancellation.
- Queue tracking.
- Notification management.
- Profile management.
- Language selection.

Doctor Module

- View appointment schedules.
- Manage patient queue.
- Update consultation status.
- Delay management.

Administrator Module

- Manage users.
- Manage doctors.

- Manage appointments.
- Generate reports.
- Configure priority rules.
- Monitor hospital performance.

Notification Module

- SMS notifications.
- Mobile push notifications.
- Appointment reminders.
- Queue alerts.

10.Non-Functional Requirements

The system should provide:

Performance

- Fast response times.
- Real-time queue updates.

Security

- Secure authentication.
- Encrypted data transmission.
- Role-based access control.

Reliability

- High system availability.
- Data backup and recovery mechanisms.

Usability

- Simple user interface.
- Easy navigation.
- Accessibility for all age groups.

Scalability

- Ability to support multiple hospitals.
- Ability to handle large user volumes.

11.Multilingual Support

One of the key features of MEDISL is multilingual accessibility.

Supported languages include:

- Sinhala (සිංහල)
- Tamil (தமிழ்)
- English

System Behavior

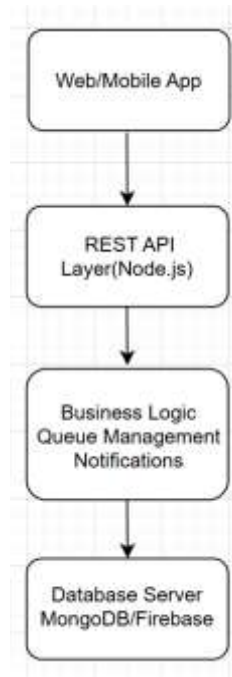
- Users select their preferred language during registration or login.
- The interface dynamically changes according to the selected language.
- Notifications are delivered in the chosen language.

Example Notification

- **English:** Your turn is next.
- **Sinhala:** ඔබගේ වාරය දුගැයි.
- **Tamil:** உங்கள் வரிசை நெருங்குகிறது.

This feature ensures inclusiveness and accessibility for all citizens.

12.System Architecture



13.Database Design

Main Entities

Patient

- Patient ID
- Name
- Contact Number
- Language Preference
- Age
- Email

Doctor

- Doctor ID
- Name
- Specialization
- Availability

Appointment

- Appointment ID
- Patient ID
- Doctor ID
- Date
- Time

Queue

- Queue ID
- Appointment ID
- Queue Number
- Status

Notification

- Notification ID
- User ID
- Message
- Delivery Status

14. Technologies Used

Component	Technology
Frontend	React.js
Mobile App	React Native
Backend	Node.js & Express.js
Database	MongoDB / Firebase
Authentication	JWT
Real-Time Updates	Socket.io / Firebase
Notifications	Firebase Cloud Messaging
SMS Service	SMS Gateway API
Version Control	Git & GitHub

15. Testing Strategy

The system will undergo several testing phases.

Functional Testing

Verifies:

- Appointment booking.
- Queue updates.
- Notifications.
- Login and registration.

Usability Testing

Verifies:

- User friendliness.
- Accessibility.
- Language switching functionality.

Performance Testing

Verifies:

- System response times.
- Concurrent user handling.

Security Testing

Verifies:

- Authentication.
- Data protection.
- Access control.

16.Expected Outcomes

The implementation of MEDISL is expected to achieve:

- Reduction in patient waiting times.
- Improved patient satisfaction.
- Better hospital resource utilization.
- Increased operational efficiency.
- Enhanced communication between patients and healthcare providers.
- Improved healthcare accessibility through multilingual support.

17.Limitations

Despite its benefits, the system has several limitations:

- Dependence on internet connectivity.
- Initial training requirements for staff.
- Integration challenges with legacy hospital systems.
- Potential resistance to technology adoption.

18.Future Enhancements

Future versions of MEDISL may include:

- AI-based waiting time prediction.
- Integration with national healthcare databases.
- Telemedicine and video consultation features.
- Voice-assisted accessibility support.
- QR-code-based patient check-in.
- Electronic medical record integration.
- Predictive analytics for hospital resource planning.

19.Feasibility Analysis

Technical Feasibility

The required technologies and development tools are widely available and well-established.

Economic Feasibility

The project can be implemented at a moderate cost while providing significant operational benefits.

Operational Feasibility

The system is designed to fit existing hospital workflows and improve service delivery.

20. Conclusion

MEDISL is an innovative healthcare solution designed to transform hospital queue and appointment management in Sri Lanka. By leveraging modern web and mobile technologies, the system addresses common challenges such as long waiting times, overcrowding, and inefficient patient management.

The integration of real-time queue tracking, multilingual support, automated notifications, and administrative monitoring will significantly improve patient experiences and healthcare service efficiency. This project has the potential to become a scalable solution that can be implemented across hospitals nationwide, contributing to the digital transformation of Sri Lanka's healthcare sector.

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